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Characterization of A-I-containing lipoproteins in subjects with A-I Milano variant

Marian C. Cheung [a], Alex V. Nichols [b], Patricia J. Blanche [b], Elaine L. Gong [b], G. Franceschini [c] and C.R. Sirtori [c]

[a] *Department of Medicine, School of Medicine, University of Washington, Seattle, WA (U.S.A.),* [b] *Donner Laboratory, Lawrence Berkeley Laboratory, University of California, Berkeley, CA (U.S.A.) and the* [c] *Center E. Grossi Paoletti, University of Milan, Milan (Italy)*

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The A-I Milano variant of apolipoprotein A-I ($A\text{-}I_M$), by virtue of its Arg-173 → Cys substitution, is capable of forming a disulfide bond with the 77-amino-acid apolipoprotein A-II polypeptide ($A\text{-}II_S$) as well as with itself to produce dimers, $A\text{-}I_M/A\text{-}II_S$ and $A\text{-}I_M/A\text{-}I_M$, respectively. A-I-containing lipoproteins (Lp): particles with A-II (Lp(A-I with A-11)) and particles without A-II (Lp(A-I without A-II)) in the plasma of two nonhyperlipidemic $A\text{-}I_M$ carriers were investigated to determine the effect of $A\text{-}I_M$ on these lipoproteins. Despite the existence of abnormal apolipoprotein dimers and the unusually low HDL cholesterol (17 and 14 mg/dl), A-I (67 and 75 mg/dl), and A-II (18 and 18 mg/dl) levels in the two carriers, the plasma A-I of the carriers was distributed between Lp(A-I with A-II) and Lp(A-I without A-II) in a proportion comparable to that observed in normals. As expected, $A\text{-}I_M/A\text{-}II_S$ mixed dimer was found in carrier Lp(A-I with A-II). However, $A\text{-}I_M/A\text{-}I_M$ dimer was located almost exclusively in carrier Lp(A-I without A-II). Chemical (dimethylsuberimidate) crosslinking of the protein moieties of the major subpopulations of Lp(A-I with A-II) and Lp(A-I without A-II) of normal and $A\text{-}I_M$ carriers showed that Lp(A-I with A-II), which is located predominantly in the 7.8–9.7 nm interval ($(HDL_{2a+3a+3b})_{gge}$), had an apparent protein molecular weight equivalent to two molecules of A-I and one to two molecules of A-II per particle. Most of the Lp(A-I without A-II) particles, located predominantly in the size intervals of 9.7–12.9 nm (designated $(HDL_{2b})_{gge}$) and 8.2–8.8 nm $(HDL_{3a})_{gge}$) had protein moieties exhibiting a molecular weight equivalence predominantly of four and three molecules of A-I, respectively. A small quantity of particles with apparent protein content of two molecules of A-I in the 7.2–8.2 nm interval ($(HDL_{3b+3c})_{gge}$) was also detected. These studies showed that in nonhyperlipidemic $A\text{-}I_M$ carriers, the occurrence of apolipoprotein dimers had not markedly affected the protein stoichiometry of Lp(A-I with A-II) and Lp(A-I without A-II).

Abbreviations: $A\text{-}I_M$, A-I Milano variant of apolipoprotein A-I; $A\text{-}II_S$, 77-amino-acid apolipoprotein A-II polypeptide; Lp, lipoproteins; apo, apolipoprotein; SDS-gPAGE, SDS-polyacrylamide gel electrophoresis on 4–30% polyacrylamide gradient gels.

Correspondence: M.C. Cheung, Northwest Lipid Research Clinic, 326 Ninth Avenue, Seattle, WA 98104, U.S.A.

Introduction

Apolipoproteins A-I and A-II are the major protein components of high-density lipoproteins (HDL). Together they constitute approx. 90% of the HDL protein mass. Apo A-I is a single polypeptide of 243 amino acids characterized by the

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absence of cysteine and isoleucine. [1] Human apo A-II is a dimer of two identical polypeptides of 77 amino acids covalently linked by a disulfide bond at the Cys-6 residue. [2] While no variant form of apo A-II has been demonstrated, several genetic variants of apo A-I have been reported [3–5]. One of these variants, A-I Milano ($A\text{-}I_M$) is characterized by a single cysteine for arginine substitution at residue 173 [6]. The presence of a cysteine molecule in $A\text{-}I_M$ results in the occurrence of $A\text{-}I_M/A\text{-}I_M$ dimers and $A\text{-}I_M/A\text{-}II_S$ ($A\text{-}II_S$: single 77-amino-acid apolipoprotein A-II polypeptide) mixed dimers in addition to apo A-I and A-II in the HDL of affected subjects. [7]

Recently, using immunochemical techniques, we isolated and characterized two types of HDL particle: particles containing both A-I and A-II (Lp(A-I with A-II)) and particles containing A-I but no A-II (Lp(A-I without A-II)). Both types of particle are heterogeneous in size. In normal subjects, Lp(A-I with A-II) contain particles of mostly three distinct sizes with mean hydrated Stokes diameters of 9.6 nm for Lp(A-I with $A\text{-}II)_1$, 8.9 nm for Lp(A-I with $A\text{-}II)_2$, and 8.0 nm for Lp(A-I with $A\text{-}II)_3$. Lp(A-I without A-II) comprise mostly particles of two distinct sizes with mean hydrated Stokes diameters of 10.8 nm for Lp(A-I without $A\text{-}II)_1$ and 8.5 nm for Lp(A-I without $A\text{-}II)_2$ [8]. Since the HDL of subjects with $A\text{-}I_M$ variant contain $A\text{-}I_M/A\text{-}I_M$ dimers as well as $A\text{-}I_M/A\text{-}II_S$ mixed dimers, the characteristics of Lp(A-I with A-II) and Lp(A-I without A-II) particles in these subjects were studied in order to determine the effect of $A\text{-}I_M$ on the structure of these two types of particle.

Methods

Subjects. Analyses were carried out in one female (13 years old, pre-puberty) and one male (29 years) normolipidemic subject with the apo $A\text{-}I_M$ variant (pedigree identification VII-48 and VII-134, respectively) [9]. Blood was collected into EDTA (1 mg/ml) after an overnight fast; the plasma was immediately air-shipped on wet ice to Berkeley. Upon removal of an aliquot for ultracentrifugation study, the remaining plasma was air-shipped to Seattle for isolation of A-I-containing lipoproteins. All lipoprotein isolation commenced within 1 week after the blood was drawn. Major clinical and biochemical data for the $A\text{-}I_M$ carriers and close relatives have been reported previously [10]. For comparison of results obtained in the present studies, similar analyses were performed on three normolipidemic female subjects (ages 39, 42 and 50 years, premenopause) without the $A\text{-}I_M$ variant. In previous studies on normal subjects and subjects with the $A\text{-}I_M$ variant, no sex and age effects were observed on HDL or A-I-containing lipoprotein subpopulations studied by gradient gel electrophoresis.

Isolation of A-I-containing lipoprotein particles. A-I-containing lipoproteins were isolated according to a previously established method [8]. To isolate lipoprotein particles containing both A-I and A-II (Lp(A-I with A-II)), 13.5 ml of plasma from the $A\text{-}I_M$ carriers, and 7 ml of plasma from the normolipidemic subjects were separately incubated with 40 ml of anti-A-II immunosorbent (capacity: 0.1 mg A-II/ml) for 1 h at 4° C. The immunosorbent was packed into an Econo-Column (Bio-Rad Laboratories, Richmond, CA) of 1.5 cm internal diameter. Proteins that did not bind to the immunosorbent were washed off with 0.01 M Tris-HCl buffer (pH 7.4)/0.15 M NaCl/1 mM EDTA/0.05% sodium azide (Tris buffer) at 30 ml/h until the A_{280} of the eluate was below 0.02 units. Lipoproteins bound to the immunosorbent were eluted with 3 M NaSCN in 0.02 M sodium phosphate buffer (pH 7.0) at 60 ml/h, and immediately dialyzed against 3 changes of 4 l Tris buffer.

To isolate lipoprotein particles containing A-I but no A-II (Lp(A-I without A-II)), plasma devoid of A-II, as determined by immunoassay (the nonbinding plasma from the above anti-A-II immunosorbent) was incubated with 15 ml of anti-A-I immunosorbent (capacity: 0.5 mg A-I/ml) for 1 h at 4° C. The nonbinding proteins and bound lipoproteins were sequentially eluted from the immunosorbent and processed as described above. Aliquots of the isolated particles were air-shipped on wet ice to Berkeley. All subsequent analyses were performed by Berkeley and/or Seattle laboratories. Unpublished observations from both laboratories showed that HDL particle sizes

studied by gradient polyacrylamide-gel electrophoresis are stable at 4°C for at least 10 days.

Gradient gel electrophoresis. Lipoprotein fractions isolated by immunoaffinity chromatography [8] or by ultracentrifugation at *d* 1.21 g/ml [11] from plasma were analyzed by nondenaturing gradient gel electrophoresis (gPAGE) by the method of Nichols et al. [12] Precast 4–30% gradient gels were used, and particle size was calibrated by high molecular weight reference proteins (Pharmacia Fine Chemicals, Piscataway, NJ). Staining, destaining, and analysis procedures were as previously described. By means of this technique, five particle size intervals have been defined previously that contain the electrophoretic peaks observed for the major HDL subpopulations [13] in plasma of normal subjects. The specific size intervals and their corresponding $(HDL)_{gge}$ designations are: 9.7–12.9 nm, $(HDL_{2b})_{gge}$; 8.8–9.7 nm, $(HDL_{2a})_{gge}$; 8.2–8.8 nm, $(HDL_{3a})_{gge}$; 7.8–8.2 nm, $(HDL_{3b})_{gge}$; and 7.2–7.8 nm, $(HDL_{3c})_{gge}$.

Fractionation of Lp(A-I with A-II) and Lp(A-I without A-II). After dialysis, the Lp(A-I with A-II) and Lp(A-I without A-II) particles isolated by immunoaffinity chromatography were concentrated to 1–4 mg protein/ml by Micro-Confilt concentrator using protein dialysis membrane of M_r = 10 000 cut-off (Biomolecular Dynamics, Beaverton, OR). 1 or 2.0 ml aliquots of the concentrated lipoprotein particles were fractionated by gel-filtration chromatography (1.5 × 95 cm) on Ultrogel AcA 34 (LKB, Bromma, Sweden) at 4°C with a flow rate of 6 ml/h. Gel-filtration fractions were analyzed by gradient gel electrophoresis [12] prior to pooling for chemical crosslinking and protein analyses.

Chemical crosslinking and analysis. Pooled fractions from the gel-filtration column were dialyzed to 100 mM sodium phosphate buffer (pH 7.2) and chemical crosslinking performed according to the method of Swaney and O'Brien [14] using dimethylsuberimidate dihydrochloride (Pierce Chemical Co., Rockford, IL). The crosslinked products were delipidated by diethyl ether/ethanol (3:1) and the molecular weights of the extracted proteins were determined by SDS-polyacryalmide gel electrophoresis on 4–30% polyacrylamide gradient gels SDS-gPAGE) (Pharmacia Fine Chemicals, Piscataway, NJ) using the method of Weber and Osborn. [15] Molecular weights of the crosslinked proteins were calculated using SDS-PAGE standards (Bio-Rad Laboratories, Richmond, CA). Chemical crosslinking and SDS-gPAGE were also performed on purified A-I in each experiment to assess the effectiveness of the crosslinking and to identify positions of the various crosslinked polymers. The completeness of crosslinking was checked using aldolase, and possible intermolecular crosslinking was evaluated by crosslinking mixtures of apo A-I and myoglobin. These tests show the conditions used to yield complete crosslinking without intermolecular artifact.

Protein analyses. Apo A-I, A-II and D contents of plasma and various subfractions were quantitated by specific radial immunodiffusion assays. [16–18] Protein composition of isolated Lp(A-I with A-II) and Lp(A-I without A-II) was studied by SDS-PAGE according to the method of Laemmli using 15% acrylamide [19]. Protein bands were visualized by staining in 0.035% Coomassie R250 dissolved in isopropanol/acetic acid/water (25:10:65) and destained in the same solvent.

Results

Plasma lipid and lipoproteins

Although it has been shown that $A\text{-}I_M$ carriers exhibit a relatively high prevalence of hypertriglyceridemia, the two subjects in this study had normal levels of plasma cholesterol and triacylglycerol (Table I) when compared (according to age and sex) to the Lipid Research Clinic Prevalence Study data. [20] In accordance with other $A\text{-}I_M$ carriers, the HDL cholesterol levels of the two subjects were extremely low. The reduction in HDL cholesterol level was accompanied by a reduction in A-I and A-II, but to a lesser extent. The plasma apo D concentrations were within the normal range. [18]

Plasma A-I is normally distributed between two populations of lipoprotein particles: particles containing A-II (Lp(A-I with A-II)), and particles in which A-II is absent (Lp(A-I without A-II)). In normolipidemic subjects with normal HDL, 27–51% of plasma A-I is found in Lp(A-I without A-II), and the A-I/A-II molar ratio in Lp(A-I with A-II) measured by our immunoassay is nearly 2 [8]. When the plasma of the $A\text{-}I_M$ carriers was

TABLE I

PLASMA LIPID AND PROTEIN CONCENTRATION IN A-I_M AND NORMAL SUBJECTS

Abbreviations: CH, cholesterol; TG, triacylglycerols; HDL-CH, high density lipoprotein-cholesterol

	Subject	Age (years)	Sex	(mg/dl)						Plasma A-I in Lp(A-I without A-II) (%)
				CH	TG	HDL-CH	A-I	A-II	D	
Milano										
	VII-48	13	F	139	61	17	67	18	5.6	53
	VII-134	29	M	155	107	14	75	18	5.9	52
Normal										
	1	42	F	182	93	60	186	47	8.6	32
	2	39	F	148	36	53	134	25	7.3	50
	3	50	F	179	32	90	208	31	8.9	50

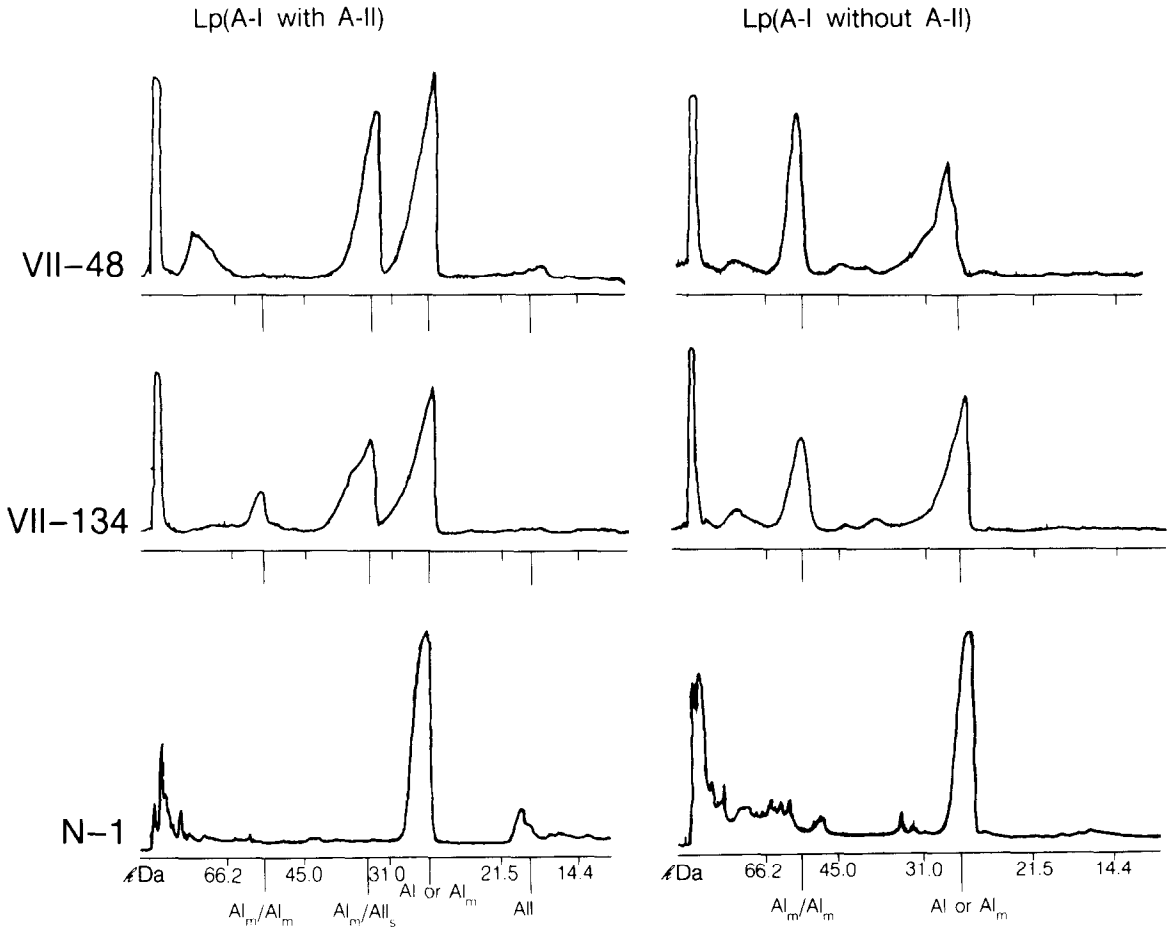


Fig. 1. SDS-gPAGE of protein moieities of Lp(A-I with A-II) and Lp(A-I without A-II) populations isolated by immunoaffinity chromatography. Carriers of A-I_M variant are identified by pedigree nomenclature (VII-48, VII-134) (Table I). Data for normal subject (N-1) are presented in Table I. Scale at bottom is calibrated according to molecular weights of a standard protein mixture run with each analysis. Molecular weights corresponding to A-I, A-I_M and A-II are indicated together with those of dimers and mixed dimers of A-I_M (in kDa).

fractionated by anti-A-II and anti-A-I immunosorbents to isolate Lp(A-I with A-II) and Lp(A-I without A-II), we found that the distribution of plasma A-I between these two populations of particles was about equal (Table I). However, unlike the Lp(A-I with A-II) of normal subjects, the apparent A-I/A-II molar ratio in the carrier Lp(A-I with A-II) particles was only slightly greater than 1. Apo D was found in both Lp(A-I with A-II) and Lp(A-I without A-II) of carriers. Approx. 38% of A-I-associated apo D was located in Lp(A-I without A-II). The distribution of apo D between the two HDL populations was comparable to our earlier findings for normal subjects [8]. No apo D was detected in the A-I-free plasma of both carriers. (The lower limit of sensitivity of our apo D immunoassay was 2 μg/ml.)

Characterization of Lp(A-I with A-II) and Lp(A-I without A-II)

By SDS-PAGE, in the absence of a reducing agent, A-I_M/A-II_S mixed dimer and A-I or A-I_M monomer were the prominent protein components detected in Lp(A-I with A-II) (Fig. 1). Interestingly, only a small quantity of A-I_M/A-I_M dimer was found in the Lp(A-I with A-II) of subject VII-134 and none was detected in this population of subject VII-48. Although the stained A-II dimer band was very weak compared to the A-I_M/A-II_S, it was not altogether clear that most A-II in Lp(A-I with A-II) was dimerized with A-I_M. Thus, when similar SDS-PAGE was performed on the Lp(A-I with A-II) population isolated from a normal subject (N-1), A-II was very poorly stained compared to A-I (Fig. 1). Given this difference in the chromogenicity of A-I and A-II in our gel system, it appeared that the intensity of staining in A-I_M/A-I_S was due mostly to A-I_M. Assuming further that the chromogenicity of A-I and A-I_M in our system was comparable, we estimate, from the densitometric scans, that an average of 40% of A-I in the carrier Lp(A-I with A-II) was dimerized with A-II_S. Combining this information with the distribution and concentration data presented in Table I, we estimate that about 4 mg/dl of plasma A-II mass was present as A-I_M/A-II_S. Analysis of SDS-PAGE of the Lp(A-I without A-II) population showed the presence of A-I and/or A-I_M as well as A-I_M/A-I_M dimer in approximately equal proportions in both carriers (Fig. 1). The protein bands at positions corresponding to A-I_M/A-I_M, A-I_M/A-II_S, and A-II could not be detected when SDS-PAGE of the particles was performed in the presence of β-mercaptoethanol (data not shown).

The size distributions of the carrier Lp(A-I with A-II) and Lp(A-I without A-II) populations were studied by gPAGE. Densitometric scans of these populations as well as corresponding populations from two normal subjects (N-1, N-2), isolated at the time the carrier plasmas were studied, are shown in Fig. 2. Comparable to the Lp(A-I with A-II) and Lp(A-I without A-II) of normal subjects, three and two major size subpopulations, respectively, could be clearly identified in the two populations isolated from carrier plasma. However, the mean particle sizes of some of the major subspecies appeared to be smaller than those of the normal subjects. In Lp(A-I with A-II), the mean hydrated Stokes diameters of the three size subspecies (Lp(A-I with A-II)$_1$, Lp(A-I with A-II)$_2$, Lp(A-I with A-II)$_3$) were 8.6 nm, 7.9 nm, and 7.7 nm for the carriers, and 9.4 nm, 8.9 nm, and 8.2 nm for the normals, respectively. In Lp(A-I without A-II), the mean hydrated Stokes diameter of the larger-size subspecies (Lp(A-I without A-II)$_1$) was 10.1 nm for the carriers and 10.4 nm for the normals; the mean hydrated Stokes diameter of the smaller-size subspecies (Lp(A-I without A-II)$_2$) was 8.5 nm for both the carriers and the normals.

Densitometric scans (Fig. 2) of the 4–30% gPAGE gels of the $d < 1.20$ g/ml plasma fraction show that, relative to the HDL distribution of the normals, the carriers have reduced area in the 9.7–12.9 nm interval ($(HDL_{2b})_{gge}$ interval) (carrier = 22%; normal = 30%) and increased area in the 7.2–8.8 nm interval ($(HDL_{3a+3b+3c})_{gge}$ interval) (carrier = 59%; normal = 48%). The ratio of the two areas for the carriers was 1.7-fold that of normal subjects.

Chemical crosslinking, using dimethylsuberimidate, was performed on total Lp(A-I with A-II) and Lp(A-I without A-II) poulations as well as on fractions generated by gel filtration of the populations. Fractions from gel filtration were pooled based on their gPAGE patterns and pools were crosslinked whenever sufficient material was available. Fig. 3 shows the SDS-gPAGE patterns

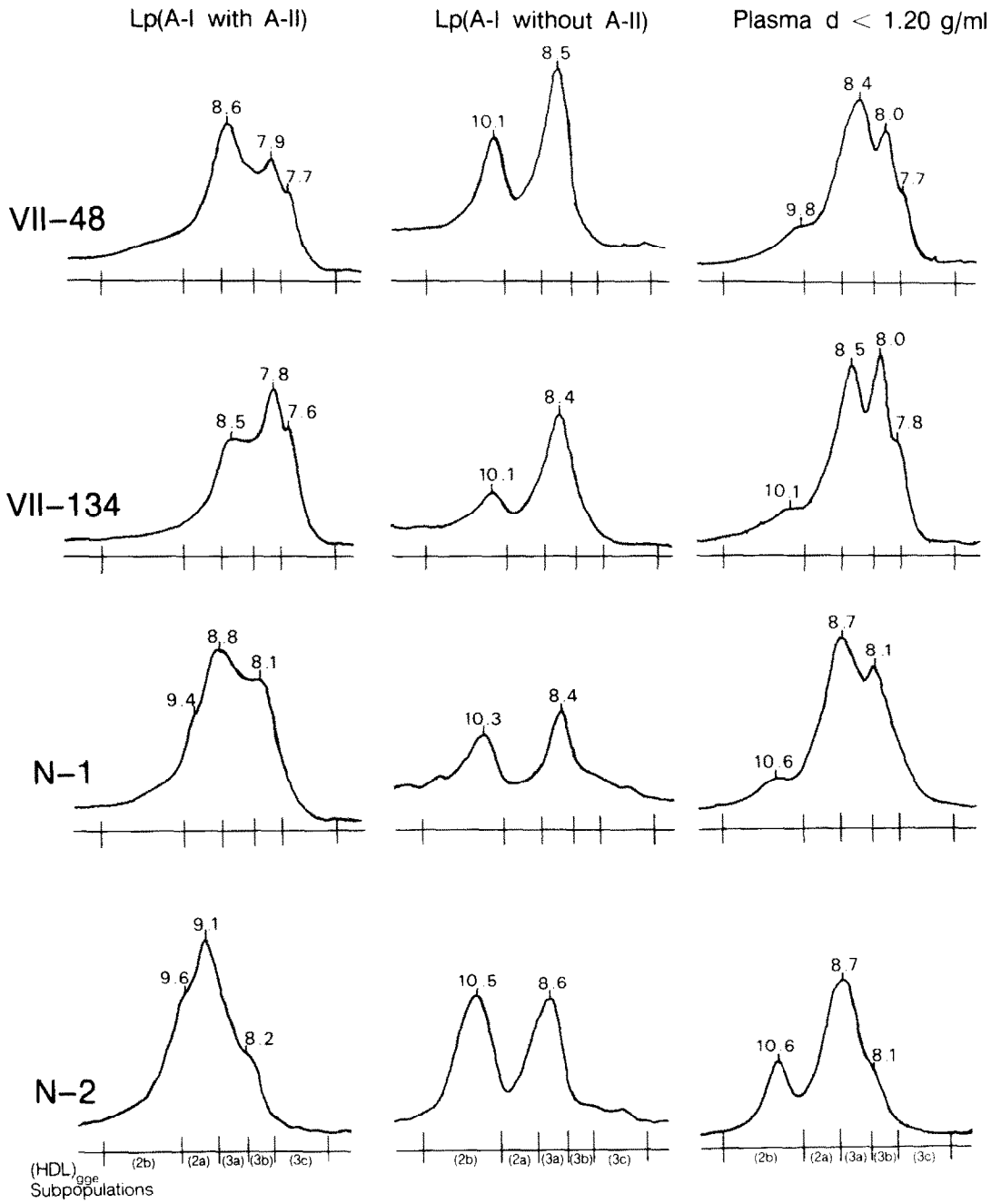
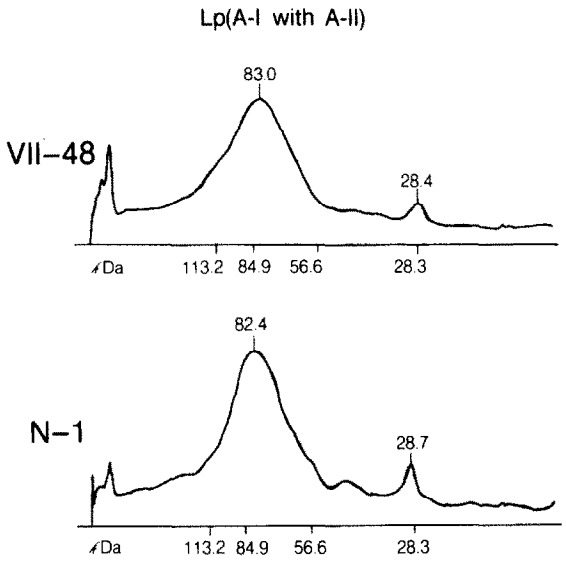


Fig. 2. gPAGE patterns of nondenatured Lp(A-I with A-II), Lp(A-I without A-II) and total $d < 1.20$ g/ml fractions isolated from plasma of carriers and normal subjects. Particle sizes of subpopulations determined at peak maxima are indicated (in nm). Scale at bottom corresponds to size intervals of major HDL subpopulations identified by gPAGE in plasma of normal subjects (see Methods).

of the total Lp(A-I with A-II) population of a carrier and a normal subject after chemical crosslinking. Similar crosslinking results (data not shown) were obtained on pools of several gel-filtration fractions from the total population; these pools exhibited, by nondenaturing gPAGE, pat-



terns with varying degrees of overlap of the major subpopulations identified previously in total Lp(A-I with A-II) populations. The combination of the overlap in particle size distributions together with the presence of A-II and possibly the C peptides complicated detailed interpretation of the crosslinking data. A broad crosslinked protein band with peak molecular weight of 82–86 kDa was detected in the total population and gel filtration fractions in all subjects studied (Fig. 3). In carriers, this range of molecular weight suggests particles containing two molecules of A-I and/or

Fig. 3. SDS-gPAGE of crosslinked protein moieties of total Lp(A-I with A-II) populations isolated from plasma of a carrier (VII-48) and normal subject (N-1). Molecular mass of crosslinked protein at predominant peak is indicated (in kDa).

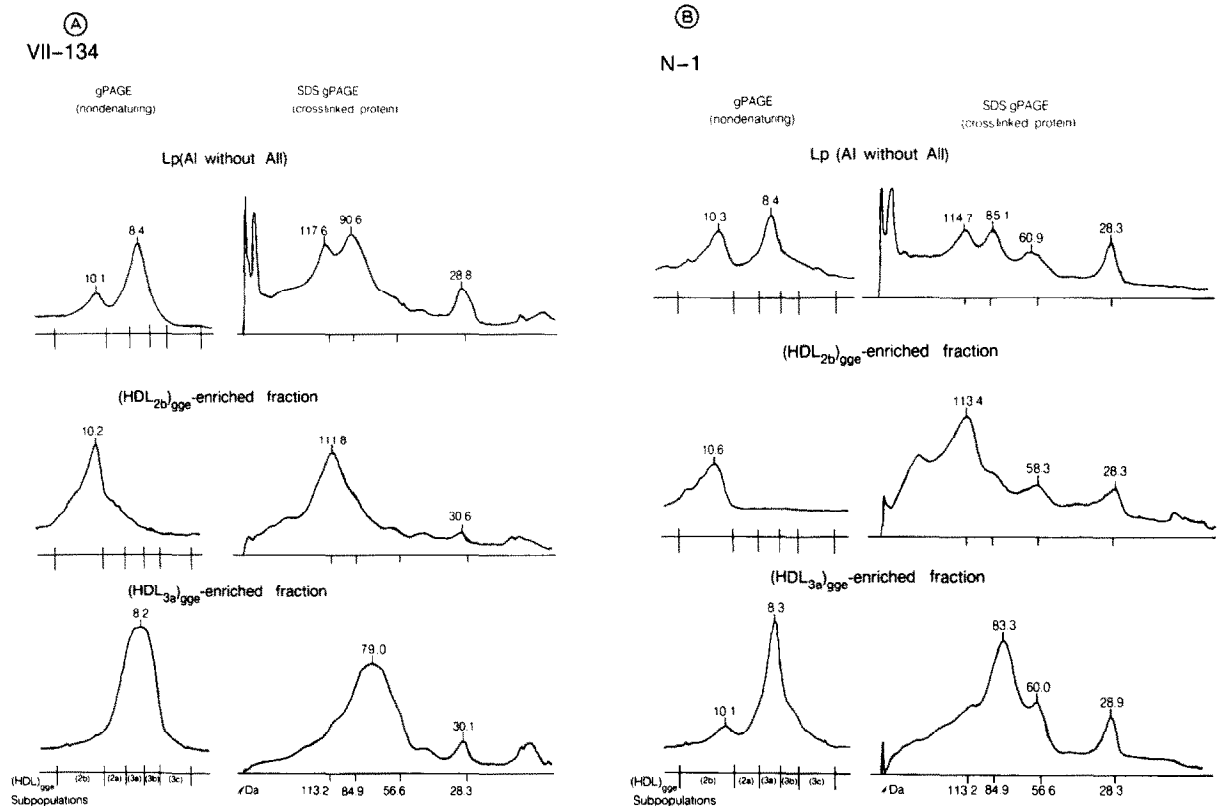


Fig. 4. SDS-gPAGE of crosslinked protein moieties of the total Lp(A-I without A-II) population and its gel-filtration fractions enriched in $(HDL_{2b})_{gge}$ and $(HDL_{3a})_{gge}$ subpopulations. Nondenaturing gPAGE patterns of the corresponding total population and gel filtration fractions prior to crosslinking are shown with particle sizes indicated (in nm). Molecular weights of crosslinked proteins at predominant peaks of SDS-gPAGE patterns are indicated (in kDa). (A) carrier (VII-134); (B) normal subject (N-1).

$A\text{-}I_M$, one molecule of A-II dimer, and one A-II half-dimer ($A\text{-}II_S$), possibly in the configuration of (A-I or $A\text{-}I_M$, $A\text{-}I_M/A\text{-}II_S$, A-II). This configuration would not apply to the normal Lp(A-I with A-II). The mean molecular weight of about 84 kDa detected in the normals is intermediate between (2A-I + 2A-II) (91.4 kDa) and (2A-I + A-II) (74.0 kDa). However, until the complete apolipoprotein composition of these particles is known, our data do not preclude the possible existence in normal subjects of Lp(A-I with A-II) in the $(HDL_{2a+3a})_{gge}$ size interval having A-I and A-II in either a 1 : 1 or 2 : 1 molar ratio or both.

Fig. 4A and B show the SDS-gPAGE patterns of the total Lp(A-I without A-II) population of a carrier (VII-134) and a normal subject (N-1), respectively, after chemical crosslinking. Prominent peaks in the patterns from both subjects occur at locations corresponding to total molecular weights equivalent to four and three molecules of A-I. Gel filtration fractions from both subjects enriched in the $(HDL_{2b})_{gge}$ subpopulation exhibit a predominance of a crosslinked protein moiety equivalent in molecular weight to four molecules of A-I. Patterns of crosslinked gel-filtration fractions enriched in the $(HDL_{3a})_{gge}$ subpopulation for both subjects indicate a predominance of protein corresponding to a molecular weight equivalence of three molecules of A-I. Our data also indicate (not shown) that Lp(A-I without A-II) particles in the $(HDL_{3c})_{gge}$ (7.2–7.8 nm) size interval occur in carriers and normals and contain proteins with a total molecular weight equivalent to two molecules of A-I. In each SDS-gPAGE pattern, peaks at molecular weight positions smaller than the prominent peak could also be observed. These peaks may represent protein moieties of smaller molecular weight particles in the pool as well as incomplete crosslinking of the proteins in the major subspecies. Peaks at the molecular weight position of monomeric A-I (especially in the normals) could be identified in each case.

Discussion

Several variant forms of A-I (A-I Giessen, Marburg, Milano, Munster) involving either deletion or substitution of an amino acid in the polypeptide chain of the most common form of A-I have been reported. [3–5] With the exception of $A\text{-}I_M$, carriers of the other variant forms of A-I do not exhibit any consistent pattern of dyslipoproteinemia. In $A\text{-}I_M$ carriers, affected kindred demonstrate strikingly low levels of HDL cholesterol and an increased prevalence of hypertriglyceridemia. [21] Their HDL is characterized by a relative reduction in HDL_2 with increased HDL_3 heterogeneity. The protein moiety of the heterogeneous HDL_3 particles is made up of A-I, $A\text{-}I_M$, A-II, $A\text{-}I_M/A\text{-}I_M$ and $A\text{-}I_M/A\text{-}II_S$ in different proportions [22]. Prompted by this observation and our previous study showing that HDL of normal subjects contains Lp(A-I with A-II) as well as Lp(A-I without A-II) particles, the present study was performed to characterize further $A\text{-}I_M$ HDL in terms of these two populations of particles.

Our success in isolating A-I-containing lipoprotein particles from the plasma of $A\text{-}I_M$ carriers using antibody to authentic A-I supports the earlier observation that $A\text{-}I_M$ is immunochemically identical to authentic A-I. [7] Using our isolation scheme, we have clearly demonstrated that the HDL of $A\text{-}I_M$ carriers is also made up of Lp(A-I with A-II) and Lp(A-I without A-II) particles in a relative proportion comparable to normal subjects. However, unlike the A-I/A-II molar ratio of 2 observed in Lp(A-I with A-II) of normal subjects, the molar ratio in carrier Lp(A-I with A-II) particles is only slightly greater than 1. This reduced A-I/A-II ratio could have resulted from an overestimation of $A\text{-}II_S$ mass by our radial immunodiffusion assay which utilized dimeric A-II with a molecular weight of 17.4 kDa purified from normal plasma as primary standard.

Analysis of the size subpopulations of carrier Lp(A-I without A-II) by nondenaturing gPAGE shows a reduced level of $Lp(A\text{-}I \text{ without } A\text{-}II)_1$, the larger and more buoyant of the two major Lp(A-I without A-II) subpopulations. [23] This is consistent with earlier observations that HDL_2 is reduced in $A\text{-}I_M$ carriers. [22] With respect to size, the mean hydrated Stokes diameter of carrier $Lp(A\text{-}I \text{ without } A\text{-}II)_1$ is approx. 0.4 nm smaller than that of the normals, while the particle size of $Lp(A\text{-}I \text{ without } A\text{-}II)_2$ is comparable to the normals. The mean hydrated Stokes diameters of $Lp(A\text{-}I \text{ with } A\text{-}II)_1$, $Lp(A\text{-}I \text{ with } A\text{-}II)_2$, and

Lp(A-I with A-II)$_3$ of normals are 9.4 nm, 8.9 nm and 8.2 nm, respectively. In carriers, the sizes of all three subpopulations are smaller: 8.6 nm, 7.9 nm and 7.7 nm, respectively. Therefore, as a whole, carrier A-I-containing lipoprotein particles are smaller than those of normals. However, due to the small number of subjects studied, it cannot be determined whether true differences in particle size exist between carriers and normals.

Previous studies of A-I_M HDL showed the presence of A-I_M/A-I_M, A-I_M/A-II_S, A-I and/or A-I_M and A-II in all HDL_3 fractions separated by gel-filtration chromatography. The percentage of A-I_M/A-I_M was found to decrease while the percentage of A-I_M/A-II_S increased from large to small particles. [22] Our present work shows that the ultracentrifugal HDL_3 subclass contains both Lp(A-I with A-II) and Lp(A-I without A-II) populations of HDL particles. SDS-gPAGE of these isolated populations provides detailed insight into the apolipoprotein distribution. In Lp(A-I without A-II), A-I and/or A-I_M and A-I_M/A-I_M are present as expected. In Lp(A-I with A-II), the A-I_M/A-II_S mixed dimer is almost exclusively observed. The near absence of A-I_M/A-I_M in Lp(A-I with A-II) is responsible for the decrease of their dimer in the smaller HDL particles in the size intervals of $(HDL_{3b})_{gge}$ and $(HDL_{3c})_{gge}$.

When the two major subpopulations of Lp(A-I without A-II) particles are separated by gel-filtration chromatography, crosslinking of the surface proteins shows that Lp(A-I without A-II)$_1$ and Lp(A-I without A-II)$_2$ of the carriers as well as of normal subjects contain proteins with a combined molecular weight equivalent to four molecules and three molecules of A-I, respectively. It is of interest to note that comparable molecular weight values of A-I-enriched fractions of HDL_2 [24] and HDL_3 [25] have been obtained in studies reported using the crosslinking methodology. In our studies, low levels of Lp(A-I without A-II) particles containing an equivalent of two molecules of A-I per particle are also detected within the $(HDL_{3c})_{gge}$ interval. Although we cannot, at present, exclude the possibility of non-A-I proteins contributing to the total observed particle protein mass, the apparent stepwise increase of A-I molecules per particle is compatible with the recent observations of Chen et al. [26] and Nichols et al. [27] on native and model HDL, respectively. These investigators showed that small spherical HDL particles from lecithin–cholesterol acyltransferase-deficient plasma [26] as well as A-I-phosphatidylcholine model complexes [27] containing two molecules of A-I are converted to particles with three molecules of A-I in the presence of the lecithine–cholesterol acyltransferase enzyme and a source of cholesterol. This increase in the number of A-I molecules per particle in the conversion of small to large particles is postulated to involve fusion of small particles during core build-up.

Chemical crosslinking of Lp(A-I with A-II) from carriers and normal subjects shows that particles in the $(HDL_{2a+3a})_{gge}$, and possibly $(HDL_{3b})_{gge}$ size intervals contain a protein moiety of molecular weight in the range of about 82–86 kDa. A molecular weight of approx. 90 kDa has been obtained by crosslinking of total HDL_3 isolated ultracentrifugally in two reported studies [22,24]. Lipoprotein concentrations in the $(HDL_{3c})_{gge}$ size interval were not sufficient for crosslinking in our studies. As described under Results, our present data do not provide a definitive apolipoprotein stoichiometry for Lp(A-I with A-II) particles from carriers or normal subjects, although our immunochemical data support an A-I/A-II molar ratio of 2.

Our present report documents that the relative mass distribution of Lp(A-I with A-II) and Lp(A-I without A-II) populations in these subjects is close to normal, notwithstanding the extremely low plasma levels of total HDL. Furthermore, the basic physicochemical characteristics of these particles within the same size intervals do not appear to differ greatly between the normolipidemic carrier and normal subjects. Since the lipid binding properties of A-I_M are somewhat different from authentic A-I [28], it is possible that the authentic A-I present in these heterogeneous A-I_M carriers is sufficient to maintain the structural integrity of the HDL particles.

The isolation of Lp(A-I with A-II) and Lp(A-I without A-II) populations by immunoaffinity chromatography and their characterization according to particle size allows us for the first time to describe the various A-I-containing lipoprotein subpopulatins according to the gPAGE classification system for total HDL [13]. In nor-

mal subjects, most Lp(A-I without A-II)$_1$ and Lp(A-I without A-II)$_2$ particles are respectively located in the $(HDL_{2b})_{gge}$ and $(HDL_{3a})_{gge}$ size intervals. Lp(A-I with A-II)$_1$ and Lp(A-I with A-II)$_3$ are found mostly in the $(HDL_{2a})_{gge}$ and $(HDL_{3b})_{gge}$ size intervals, respectively. However, Lp(A-I with A-II)$_2$ appears to be a mixture of species within the $(HDL_{2a})_{gge}$ and $(HDL_{3a})_{gge}$ size intervals. Gradient PAGE analysis of total HDL prepared by ultracentrifugation does not differentiate between Lp(A-I without A-II)$_2$ and Lp(A-I with A-II)$_2$ subpopulations (Fig. 2).

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